

EXPERIMENT 7:

AIM:

To implement a simple stochastic part-of-speech tagging algorithm using a basic probabilistic model in Python.

PROGRAM:

```
import random
```

```
pos_tags = {
```

```
    "dog": ["NN"],
```

```
    "cat": ["NN"],
```

```
    "runs": ["VB"],
```

```
    "eats": ["VB"],
```

```
    "fast": ["RB"],
```

```
    "quickly": ["RB"]
```

```
}
```

```
sentence = input("Enter a sentence: ")
```

```
words = sentence.split()
```

```
for word in words:
```

```
    if word in pos_tags:
```

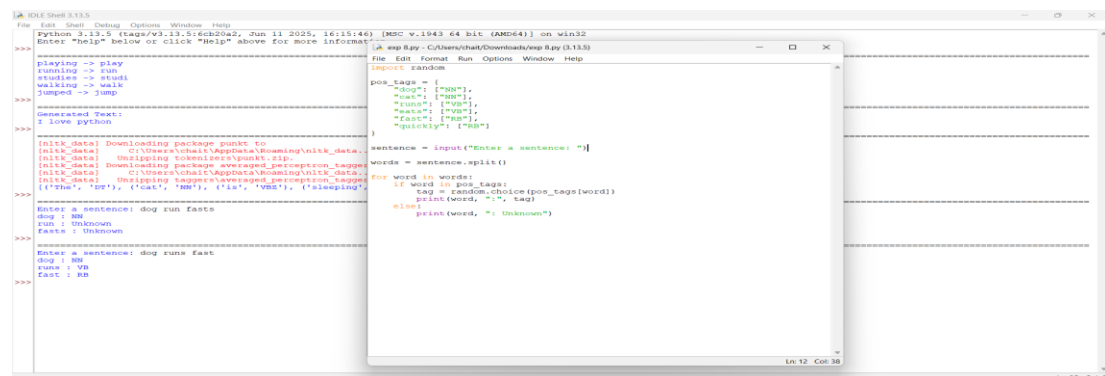
```
        tag = random.choice(pos_tags[word])
```

```
        print(word, ":", tag)
```

```
    else:
```

```
        print(word, ": Unknown")
```

OUTPUT:



```
File Edit Shell Debug Options Window Help
python 3.13.5 (tags/v9.13.5:6b20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information
>>>
playing -> play
studying -> run
studies -> studi
walking -> walk
jumped -> jump
=====Text=====
I love python
[nltk_data] Downloading package punkt to
[nltk_data] C:\Users\chait\AppData\Localing\nltk_data\
[nltk_data] Unzipping tokenizers\punkt.zip
[nltk_data] Downloading package averaged_perceptron_tagger
[nltk_data] C:\Users\chait\AppData\Localing\nltk_data\
[nltk_data] Unzipping taggers\averaged_perceptron_tagger
[{"The", "DT"}, {"cat", "NN"}, {"is", "VBZ"}, {"sleeping",
Enter a sentence: dog run fast
dog : NN
run : Unknown
fast : Unknown
Enter a sentence: dog runs fast
dog : NN
runs : VB
fast : RB
>>>
```

```
File Edit Format Run Options Window Help
exp 8.py - C:\Users\chait\Downloads\exp 8.py (0.13.5)
Import random
pos_tags = {
    "dog": ["NN"],
    "cat": ["NN"],
    "runs": ["VB"],
    "eats": ["VB"],
    "fast": ["RB"],
    "quickly": ["RB"]
}
sentence = input("Enter a sentence: ")
words = sentence.split()
for word in words:
    if word in pos_tags:
        tag = random.choice(pos_tags[word])
        print(word, ":", tag)
    else:
        print(word, ": Unknown")
Ln 12 Col 38
Ln 35 Col 0
```

